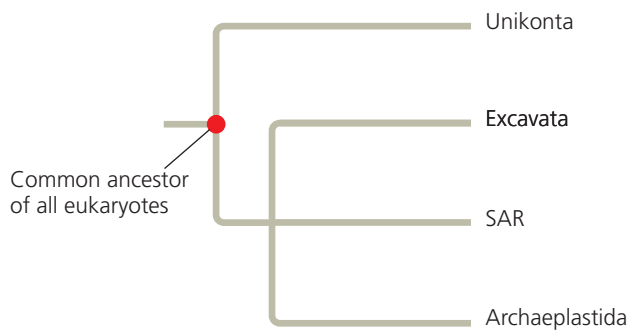


▼ Figure 28.26 Inquiry

What is the root of the eukaryotic tree?

Experiment Responding to the difficulty in determining the root of the eukaryotic phylogenetic tree, Romain Derelle and colleagues used a new approach. They studied 39 nuclear genes that eukaryotes acquired by horizontal gene transfer from a single bacterial source: the alpha proteobacterium endosymbiont that gave rise to the mitochondrion. These 39 genes were known to be orthologous (see Figure 26.18). As a result, differences among these genes reflect the history of speciation events that have occurred since the eukaryotes began to diverge from their last common ancestor—making these genes well suited for inferring the root of the eukaryotic tree. Derelle and colleagues obtained the sequences of these genes for 67 species of eukaryotes representing a wide range of eukaryotic taxa. Those sequences were then used to infer evolutionary relationships among the four eukaryotic supergroups shown in Figure 28.5.

Results



Conclusion The results show that Excavata, SAR, and Archaeplastida form a clade, which supports the hypothesis that the root of the tree is located between the unikonts and all other eukaryotes.

Data from R. Derelle et al., Bacterial proteins pinpoint a single eukaryotic root, *Proceedings of the National Academy of Sciences USA* 112:E693–699 (2015).

WHAT IF? Other recent evidence indicates that the root of the eukaryotic tree may lie between a clade that includes unikonts and excavates and all other eukaryotes. Draw the tree suggested by this result.

donors (Figure 28.26). Results from this approach indicate that Excavata, SAR, and Archaeplastida share a more recent common ancestor than any of them does with Unikonta. This suggests that the root of the tree is located between the unikonts and all other eukaryotes, which implies that the unikonts were the first eukaryotic supergroup to diverge from all other eukaryotes. This idea remains controversial and will require more supporting evidence to be widely accepted.

Amoebozoans

The **amoebozoan** clade includes many species of amoebas that have lobe- or tube-shaped pseudopodia rather than the threadlike pseudopodia found in rhizarians. Amoebozoans include tubulinids, slime molds, and entamoebas.

Tubulinids

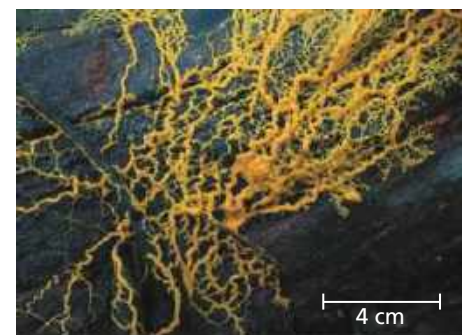
Tubulinids constitute a large and varied group of amoebozoans that have lobe- or tube-shaped pseudopodia. These unicellular protists are ubiquitous in soil as well as freshwater and marine environments. Most are heterotrophs that actively seek and consume bacteria and other protists; one such tubulinid species, *Amoeba proteus*, is shown in Figure 28.5. Some tubulinids also feed on detritus (nonliving organic matter).

Slime Molds

Slime molds, or mycetozoans (from the Latin, meaning “fungus animals”), once were thought to be fungi because, like fungi, they produce fruiting bodies that aid in spore dispersal. However, DNA sequence analyses indicate that the resemblance between slime molds and fungi is a case of evolutionary convergence. Slime molds have diverged into two main branches, plasmodial slime molds and cellular slime molds. We’ll compare their characteristics and life cycles.

Plasmodial Slime Molds Many plasmodial slime molds are brightly colored, often yellow or orange (Figure 28.27). As they grow, they form a mass called a plasmodium, which can be many centimeters in diameter. (Don’t confuse a slime mold’s plasmodium with the genus *Plasmodium*, which includes the parasitic apicomplexan that causes malaria.) Despite its size, the plasmodium is not multicellular; it is a single mass of cytoplasm that is undivided by plasma membranes and that contains many nuclei. This “supercell” is the product of mitotic nuclear divisions that are not followed by cytokinesis. The plasmodium extends pseudopodia through moist soil, leaf mulch, or rotting logs, engulfing food particles by phagocytosis as it grows. If the habitat begins to dry up or there is no food left, the plasmodium stops growing and differentiates into fruiting bodies that function in sexual reproduction.

► **Figure 28.27**
A mature plasmodium, the feeding stage in the life cycle of a plasmodial slime mold.



Cellular Slime Molds The life cycle of the protists called cellular slime molds can prompt us to question what it means to be an individual organism. The feeding stage of these organisms consists of solitary cells that function individually, but when food is depleted, the cells form a slug-like aggregate that functions as a unit (Figure 28.28). Unlike the feeding stage (plasmodium) of a plasmodial slime mold, these aggregated cells remain separated by their individual plasma membranes. Ultimately, the aggregated cells form an asexual fruiting body.